

Solution Requirements

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	No user login is required; the application is open-access for demonstration purposes.	Low
2	Authorization levels	Single-role access; there are no differentiated user permission levels in this prototype.	Low
3	External interfaces	No external interfaces currently; the design allows future integration with a live weather API (e.g., OpenWeatherMap).	Medium
4	Transactions processing	The user submits weather parameters via an HTML form; Flask processes the request and returns a prediction synchronously.	High
5	Reporting	Displays a single instant prediction result (Flood / No Flood) on the same page after submission.	Medium
6	Business rules, etc.	A prediction is generated only when all 10 required numeric weather fields are provided.	High
7	Compliance to laws or regulations	No personal data is collected or stored, so no specific data-privacy regulation applies at this stage.	Low
8	Other	The model can be retrained on updated datasets via train_model.py without changing the web application code.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The prediction should be returned almost instantly after form submission.	Response time < 2 seconds
2	Scalability	The lightweight Flask app and pre-trained model should support multiple simultaneous demo users.	Supports 50+ concurrent users on standard hosting
3	Security & Data Privacy	No sensitive personal data is stored; inputs are used only for the active prediction request.	No persistent storage of user inputs
4	Reliability & Availability	The application should run without crashing for valid numeric inputs.	99% uptime during the evaluation/demo period
5	Usability & Accessibility	A simple single-page HTML form should be usable without training.	First-time user completes a prediction in < 1 minute
6	Other	The codebase should remain easy to maintain and extend with new features.	Modular functions kept in predict.py and

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
			train_model.py